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New Family of Iterative Methods Based on the Ermakov–Kalitkin Scheme for Solving Nonlinear Systems of Equations

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Abstract—A new one-parameter family of iterative methods for solving nonlinear equations and systems is constructed. It is proved that their order of convergence is three for both equations and systems. An analysis of the dynamical behavior of the methods shows that they have a larger domain of convergence than previously known iterative schemes of the second to fourth orders. Numerical results suggest that the methods are also preferable in terms of their relative stability and the number of iteration steps. The methods are compared with previously known techniques as applied to a system of two nonlinear equations describing the dynamics of a passively gravitating mass in the Newtonian circular restricted four-body problem formulated on the basis of Lagrange's triangular solutions to the three-body problem.

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1. INTRODUCTION

The task addressed in this paper is to solve a nonlinear equation or a system of equations of the form

$$f(x) = 0, \quad f: D \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^n, \quad n \geq 1 \tag{1}$$

by applying iterative methods. Problem (1) is a prototype for a huge number of nonlinear problems solved by numerical methods. Numerous multistep iterative methods of various orders have been designed for solving a variety of problems in the last decade (see, e.g., [1–4]). Nevertheless, the simple and universal Newton method with a quadratic convergence rate in a sufficiently close neighborhood of a root remains the most popular and efficient iterative technique. However, many important applications in various fields of science and engineering are described by essentially nonlinear systems and a "good" initial approximation is frequently required for Newton's method to converge. In this case, a key property is that a method converges when the initial approximation is poor, i.e., far from the root, in which case Newton's method does not work. Considerable progress in this area was achieved by Soviet computational mathematicians.

Intended for enlarging the domain of convergence, various efficient techniques for step size control in Newton's method were proposed in [5–7]. In the general case, a damped modification of Newton's method has the form

$$x_{k+1} = x_k - \beta_k[f'(x_k)]^{-1}f(x_k), \quad k = 0, 1, \ldots, \tag{2}$$

where β_k is a sequence of real numbers determined by a certain formula or algorithm and $f'(x_k)$ denotes the derivative for $n = 1$ or the Jacobian for $n > 1$. Specifically, when $\beta_k = 1$, iterative scheme (2) becomes the classical Newton method.

In [5] Kalitkin and Ermakov specified β_k as

$$\beta_k = \frac{\|f(x_k)\|^2}{\|f(x_k)\|^2 + \|f(x_k - [f'(x_k)]^{-1}f(x_k))\|^2}, \tag{3}$$

where $\|\cdot\|$ denotes any norm. Iterative scheme (2) with the coefficient β_k given by (3) is a fairly efficient method, which will be illustrated below by examples.

In this paper, we propose a one-parameter family of two-step iterative methods for solving systems (1). A damped Newton method is used at the first step (as a predictor). The second step (corrector) resembles Ermakov–Kalitkin scheme (2), (3) and represents an iterative procedure involving three functions evaluations at every iteration. For equations ($n = 1$), the iterative procedure is

$$\begin{aligned} y_k &= x_k - \alpha\frac{f(x_k)}{f'(x_k)}, \\ x_{k+1} &= y_k - \frac{f(x_k)^2}{bf(x_k)^2 + cf(y_k)^2}\frac{f(y_k)}{f'(x_k)}, \quad k = 0, 1, \ldots, \end{aligned} \tag{4}$$

where α, b, and c are parameters. For systems of equations ($n > 1$), the iterative scheme is

$$\begin{aligned} y^{(k)} &= x^{(k)} - \alpha[f'(x^{(k)})]^{-1}f(x^{(k)}), \quad k = 0, 1, \ldots, \\ x^{(k+1)} &= y^{(k)} - \left[bI + c\alpha^2\left(\frac{1}{\alpha}I - [f'(x^{(k)})]^{-1}[y^{(k)}, x^{(k)}; f]\right)^2\right]^{-1}[f'(x^{(k)})]^{-1}f(y^{(k)}), \end{aligned} \tag{5}$$

where $[y^{(k)}, x^{(k)}; f]$ denotes the divided differences of the operator f with respect to $y^{(k)}$ and $x^{(k)}$. The identity matrix is denoted by I.

Below, we prove that the proposed method (referred to as PM) has the third order of convergence both in the one-dimensional case and for systems of equations with constrained parameters α, b, and c.

PM is compared with well-known second-order techniques, such as Newton's method and the Ermakov–Kalitkin method (2), (3). Additionally, we use Traub's third-order method [8] with the iterative scheme given by

$$\begin{aligned} y_k &= x_k - [f'(x_k)]^{-1}f(x_k), \\ x_{k+1} &= x_k - [f'(x_k)]^{-1}(f(x_k) + f(y_k)), \quad k = 0, 1, \ldots \end{aligned} \tag{6}$$

and Jarratt's fourth-order method [9] with the iterative procedure

$$\begin{aligned} z_k &= x_k - \frac{2}{3}[f'(x_k)]^{-1}f(x_k), \\ x_{k+1} &= x_k - \frac{1}{2}[3f'(z_k) - f'(x_k)]^{-1}(3f'(z_k) + f'(x_k))[f'(x_k)]^{-1}f(x_k), \quad k = 0, 1, \ldots. \end{aligned} \tag{7}$$

Comparing these five methods, we construct their basins of attraction in the phase plane. The necessary concepts and definitions from the theory of holomorphic dynamical systems are given in Subsection 2.2.

The basic material of this paper is presented in two sections. Specifically, equations ($n = 1$) are addressed in Section 2, and systems of equations ($n > 1$), in Section 3. To test the methods in the multidimensional case ($n > 1$), we use a system of two nonlinear equations describing the motion of bodies in celestial mechanics. This system determines the equilibrium position of a body with a "zero" mass in the gravitational field generated by three massive bodies that form a Lagrange triangle [10, 11]. The system has two parameters and, depending on their values, has 8 to 10 solutions. Accordingly, the study of basins of attraction in the phase plane for this system is of interest in itself. Below, we show that method (5) improves the basins of attraction, making their boundaries more "uniform" and "regular" with less "noise" or "chaos."

2. STUDY OF THE PM IN THE CASE OF AN EQUATION

First, the local order of convergence of the PM family is analyzed in the scalar case. Then the methods are compared in terms of dynamics as applied to particular equations. At the end of this section, we present numerical results and discuss computational characteristics, such as the initial approximation, the number of iterations, and the residual norm. Additionally, the theoretical and computed orders of convergence of the methods are compared.

2.1. Convergence Analysis

Below, we find the values of b and c depending on α for which method (4) has the third order of convergence.

Theorem 1. *Let $\xi \in I$ be a simple zero of a differentiable function $f: I \subset \mathbb{R} \longrightarrow \mathbb{R}$ in the open interval I, and let x_0 be an initial approximation close to ξ. Then method* (4) *has the third order of convergence if $b = \frac{1+\alpha^2}{2\alpha^2}$ and $c = \frac{1+\alpha}{2(\alpha-1)\alpha^2}$, where $\alpha \notin \{0, 1\}$. Moreover, the error equation for this method has the form*

$$e_{k+1} = \left(-\frac{1}{2}(-2+2\alpha+\alpha^2+\alpha^3)c_2^2 + 2(-1+\alpha)c_3\right)e_k^3 + O(e_k^4),$$

where $c_k = (1/k!)\frac{f^{(k)}(\xi)}{f'(\xi)}$ for $k = 1, 2, \ldots$ and $e_k = x_k - \xi$.

Proof. It is based on Taylor series expansions of the elements involved in iteration expression (4).

Performing Taylor expansions about ξ, we have

$$f(x_k) = f'(\xi)[e_k + c_2e_k^2 + c_3e_k^3] + O(e_k^4)$$

and

$$f'(x_k) = f'(\xi)[1 + 2c_2e_k + 3c_3e_k^2 + 4c_4e_k^3] + O(e_k^4).$$

Substituting these expressions into the first step of scheme (4) yields

$$y_k - \xi = (1-\alpha)e_k + \alpha c_2e_k^2 + 2\alpha(-c_2^2 + c_3)e_k^3 + O(e_k^4).$$

Then, expanding $f(y_k)$ in a Taylor series around ξ, we find

$$f(y_k) = f'(\xi)[(1-\alpha)e_k + (1-\alpha+\alpha^2)c_2e_k^2 - (2\alpha^2c_2^2 + (-1+\alpha-3\alpha^2+\alpha^3)c_3)e_k^3] + O(e_k^4).$$

Combining these expressions gives

$$\begin{aligned} u = \frac{f(x_k)^2}{bf(x_k)^2 + cf(y_k)^2} &= \frac{1}{b+(-1+\alpha)^2c} + \frac{2c(-1+\alpha)\alpha^2c_2}{(b+(-1+\alpha)^2c)^2}e_k \\ &+ \frac{\alpha^2c((-(-6+6\alpha+\alpha^2)b + 3(-1+\alpha)^2(2-2\alpha+\alpha^2)c)c_2^2)}{(b+(-1+\alpha)^2c)^3} \\ &- \frac{\alpha^2c(2(3-4\alpha+\alpha^2)(b+(-1+\alpha)^2c)c_3)}{(b+(-1+\alpha)^2c)^3}e_k^2 + O(e_k^3). \end{aligned}$$

As a result, the error at the last iteration is given by

$$x_{k+1} - \xi = y_k - \xi - \frac{f(x_k)^2}{bf(x_k)^2 + cf(y_k)^2}\frac{f(y_k)}{f'(x_k)} = -\frac{(-1+\alpha)(-1+b+(-1+\alpha)^2c)}{b+(-1+\alpha)^2c}e_k$$

$$+ \frac{(\alpha b^2 + (-1+\alpha)^2c(1+\alpha^2(1-2c)+\alpha(-1+c)+\alpha^3c)}{(b+(-1+\alpha)^2c)^2}e_k^2 + \frac{b(1+2\alpha^3c+\alpha(-1+2c)-\alpha^2(1+4c))}{(b+(-1+\alpha)^2c)^2}c_2e_k^2$$

$$+ \left(-2\alpha - \frac{2(-1+\alpha)\alpha^2(-1+\alpha+\alpha^2)c}{(b+(-1+\alpha)^2c)^2} + \frac{(1+\alpha)(-2+4\alpha)}{b+(-1+\alpha)^2c}\right.$$

$$\left. + \frac{(-1+\alpha)a^2c((6-6\alpha-\alpha^2)b + 3(-1+\alpha)^2(2-2\alpha+\alpha^2)c)}{(b+(-1+\alpha)^2c)^3}\right)c_2^2e_k^3$$

$$+\left(2\alpha-\frac{2(-1+\alpha)\alpha^2(3-4\alpha+\alpha^2)c}{(b+(-1+\alpha)^2c)^2}+\frac{(1+\alpha)(2-4\alpha+\alpha^2)}{b+(-1+\alpha)^2c}\right)c_3e_k^3+O(e_k^4).$$

Now three coefficients e_k, e_k^2, and $c_2e_k^2$ form a system with unknowns b and c. This system can have two solutions: $\alpha = b = 1$ or $b = \frac{1+\alpha^2}{2\alpha^2}$, $c = \frac{1+\alpha}{2(\alpha-1)\alpha^2}$. However, the first solution is dropped, since some of the denominators in the Taylor expansion vanish in that case. Summarizing, the family of methods (4) has the third order of convergence and the error equation is given by

$$e_{k+1} = \left(-\frac{1}{2}(-2+2\alpha+\alpha^2+\alpha^3)c_2^2+2(-1+\alpha)c_3\right)e_k^3+O(e_k^4).$$

2.2. Dynamic Behavior of the Methods

Below, we construct phase planes for the family of methods (4) and for certain classical methods as applied to the following equations:

$$f_1(x) = \arctan(x), \quad \xi = 0,$$

$$f_2(x) = \arctan(x)-\frac{2x}{1+x^2}, \quad \xi_1 = 0, \quad \xi_2 \approx -1.3917, \quad \xi_3 \approx 1.3917,$$

$$f_3(x) = \frac{x^2-1}{x^2+1}+1, \quad \xi = 0.$$

Before analyzing the dynamic behavior of the iterative methods, we recall some basic concepts and definitions from the theory of holomorphic dynamical systems (for more details, see, e.g., [12, 13]).

Let $R: \hat{\mathbb{C}} \longrightarrow \hat{\mathbb{C}}$ be the operator determining the result of applying an iterative method to a particular function, where $\hat{\mathbb{C}}$ is the Riemann sphere. The orbit of a point $z_0 \in \hat{\mathbb{C}}$ is defined as a subset of the phase space consisting of consecutive images of z_0 under the operator R and has the form $\{z_0, R(z_0), \ldots, R^n(z_0), \ldots\}$.

The dynamical behavior of a fixed point of R in the complex plane can be classified depending on its asymptotics. The point $z_0 \in \mathbb{C}$ is a fixed point of R if $R(z_0) = z_0$. A fixed point is attracting, repelling, or neutral if $|R'(z_0)|$ it is less than, greater than, or equal to 1, respectively. Moreover, if $|R'(z_0)| = 0$, the fixed point is said to be superattracting.

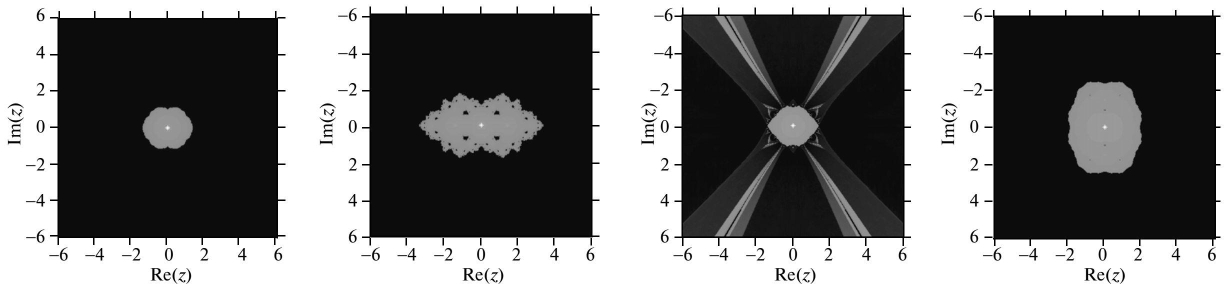
If z_f^* is an attracting fixed point of R, then its basin of attraction $\mathcal{A}(z_f^*)$ is defined as a set of n-preimages such that

$$\mathcal{A}(z_f^*) = \{z_0 \in \hat{\mathbb{C}} : R^n(z_0) \longrightarrow z_f^*, n \longrightarrow \infty\}.$$

The union of all basins of attraction in the phase plane forms the Fatou set $\mathcal{F}(R)$. The Julia set $\mathcal{J}(R)$ is the complement of the Fatou set in $\hat{\mathbb{C}}$ and establishes the boundaries between the basins of attraction.

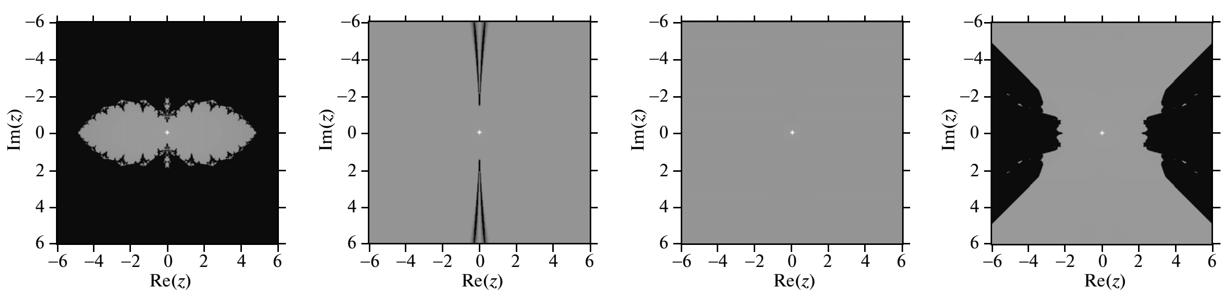
The basins of attraction of the PM and classical methods are displayed using the software described in [14]. For this purpose, we introduce a grid with a 400-point step in each axis. Each grid point is used as a different initial approximation for all the methods under comparison For the equations with $f_1(x)$ and $f_3(x)$, each having a single root, the point is shown by orange if the solution is achieved after less than 80 iteration steps. In the case of $f_2(x)$, when there are three solutions, the second and third basins are depicted in green and blue. The brighter the color, the fewer iterations are requited for achieving the root. If the method does not converge after 80 iteration steps, the corresponding point is depicted in black.

Several representatives of the PM class (with $\alpha = -0.45, -0.01, 0.1, 0.45$) were compared with previously known iterative procedures of various orders (namely, Newton's, Ermakov–Kalitkin, Traub's, and Jarratt's). Figure 1 shows their basins of attraction in the case of $f_1(x)$. Note that the largest basins in the complex plane have the Ermakov–Kalitkin and Jarratt methods (see Figs. 1b and 1d). Taking into account only real numbers, we conclude that the Ermakov–Kalitkin method converges on approximately [–3.5, 3.5]. The



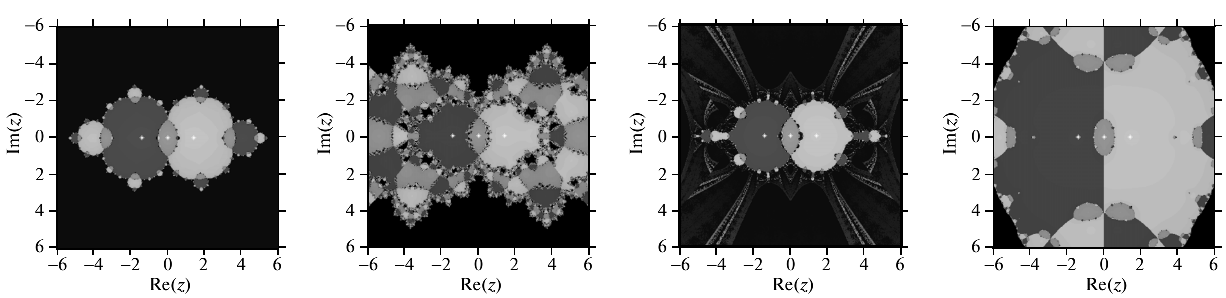
(a) Newton (b) Ermakov–Kalitkin (c) Traub (d) Jarratt

Fig. 1. Phase planes for previously known methods as applied to the equation with $f_1(x)$.



(a) $\alpha = -0.45$ (b) $\alpha = -0.01$ (c) $\alpha = 0.1$ (d) $\alpha = 0.45$

Fig. 2. Phase planes for the family of methods (4) as applied to the equation with $f_1(x)$.



(a) Newton (b) Ermakov–Kalitkin (c) Traub (d) Jarratt

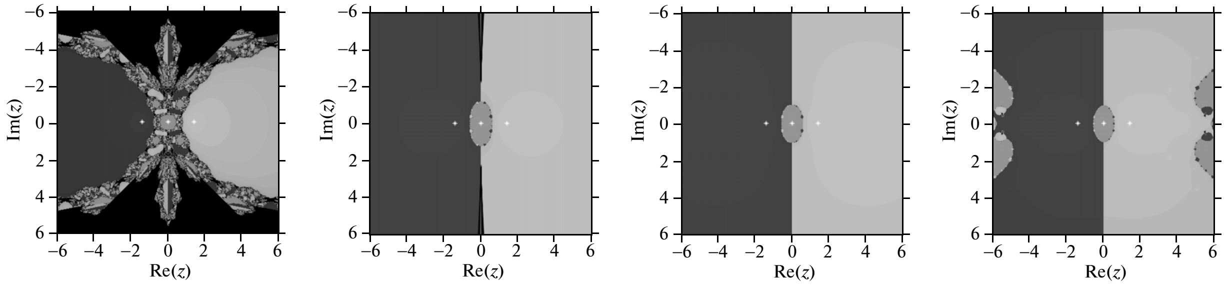
Fig. 3. Phase planes for previously known methods as applied to the equation with $f_2(x)$.

Traub (6) and Newton schemes converge roughly identically, namely, for all initial approximations from the interval $[-1.8, 1.8]$, while Jarratt method (7) has a slightly larger domain of convergence $[-2, 2]$.

An analysis of the dynamic behavior of PM shows that the largest domains of convergence occur when α is close to zero (Figs. 2b and 2c) with a real interval of convergence longer than $[-8, 8]$.

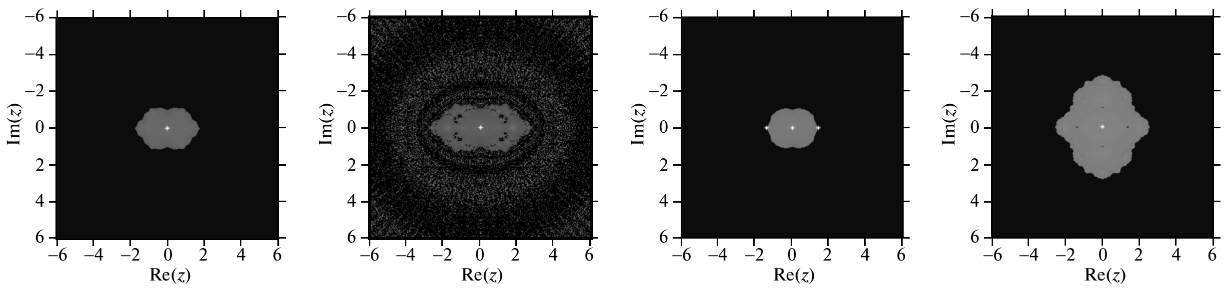
The dynamic behavior of the considered methods as applied to the equation with $f_2(x)$ is presented in Figs. 3 and 4. Once again, Newton's and Traub's methods have a smaller domain of convergence. The basins of attraction in the phase plane for the Ermakov–Kalitkin method are rather chaotic and complex in shape, although the method itself has a larger domain of convergence. The dynamics of PM is shown in Fig. 4, where, for α close to zero, PM is the most stable of all considered methods. Moreover, for the equation with $f_2(x)$, PM (for $\alpha = 0.1$, see Fig. 4c) has a rather large domain of convergence for real initial approximations (al least $[-26, 26]$).

Now consider the function $f_3(x) = \dfrac{x^2 - 1}{x^2 + 1} + 1$ and examine the dynamics of the above methods. The basins of attraction of the unique root $\xi = 0$ are presented for various classical methods (see Fig. 5) and for



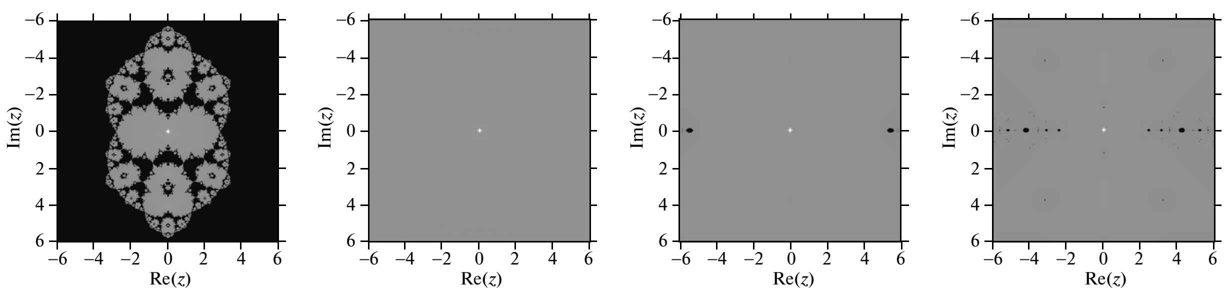
(a) $\alpha = -0.45$ (b) $\alpha = -0.01$ (c) $\alpha = 0.1$ (d) $\alpha = 0.45$

Fig. 4. Phase planes for the family of methods (4) as applied to the equation with $f_2(x)$.



(a) Newton (b) Ermakov–Kalitkin (c) Traub (d) Jarratt

Fig. 5. Phase planes for previously known methods as applied to the equation with $f_3(x)$.



(a) $\alpha = -0.45$ (b) $\alpha = -0.01$ (c) $\alpha = 0.1$ (d) $\alpha = 0.45$

Fig. 6. Phase planes for the family of methods (4) as applied to the equation with $f_3(x)$.

the PM family with various α (see Fig. 6). The classical methods hardly converge on the real interval [−3, 3], whereas PM with $\alpha = -0.01$ converges for all initial approximations from [−140, 140]. Once again, PM produces the best results for small α (in absolute value). Thus, the dynamics of PM presented in Figs. 6b and 6c suggest that this method is more stable.

2.3. Numerical Tests

The computations were performed in Mathematica 9 with long arithmetics and 10000 significant digits. In all the examples, the iterative procedures were terminated when the accuracy of the root in terms of the function residual reached 10^{-2000}.

Below are the results of a numerical experiment in which the Newton, Ermakov–Kalitkin, Traub, and Jarratt methods were compared to PM with $\alpha = 0.1$. We used the same equations with $f_1(x)$, $f_2(x)$, and $f_3(x)$ as in Subsection 2.2. All the results are given in Table 1, which demonstrates the relation between the ini-

Table 1. Numerical results for the equations

Characteristic	Method	Equation								
		$f_1(x)$			$f_2(x)$			$f_3(x)$		
		$x_0 = 1.1$	$x_0 = 3.2$	$x_0 = 7.2$	$x_0 = 2.8$	$x_0 = 5.8$	$x_0 = 24$	$x_0 = 0.3$	$x_0 = 1.6$	$x_0 = 4.8$
Iteration	N	9	–	–	12	–	–	3320	3322	–
	E–K	7	10	–	10	8	–	3618	3617	–
	PM	7	8	8	7	7	7	2094	2096	2098
	T	6	–	–	7	–	–	2346	–	–
	J	5	–	–	6	6	–	1660	–	–
ACOC	N	3	–	–	2	–	–	1	1	–
	E–K	3	3	–	2	3	–	1	1	–
	PM	3	3	3	3	3	3	1	1	1
	T	5	–	–	3	–	–	1	–	–
	J	5	–	–	4	4	–	1	–	–
$\|f(x_k)\|$	N	1e-4577	–	–	1e-2427	–	–	1e-2001	1e-2001	–
	E–K	1e-2855	1e-5763	–	1e-2081	1e-4446	–	1e-2001	1e-2001	–
	PM	1e-2561	1e-5422	1e-4248	1e-3472	1e-2553	1e-3368	1e-2001	1e-2001	1e-2001
	T	1e-3580	–	–	1e-2209	–	–	1e-2001	–	–
	J	1e-3163	–	–	1e-7888	1e-1889	–	1e-2001	–	–

tial approximations, the number of iterations, and the residual norm of the function for the various equations. Additionally, Table 1 gives the approximated computational order of convergence (ACOC) proposed in [15]. Serving as an auxiliary tool for evaluating the convergence rate of the compared iterative methods in practice, the ACOC index is defined by the formula

$$p \approx \text{ACOC} = \frac{\ln(|x_{k+1} - x_k|/|x_k - x_{k-1}|)}{\ln(|x_k - x_{k-1}|/|x_{k-1} - x_{k-2}|)}.$$

Let us explain the values of ACOC presented in Table 1. The error equations for the Newton (N), Ermakov–Kalitkin (E–K), Traub (T), and Jarratt (J) methods depend only on $c_2 = \frac{1}{2}\frac{f''(\xi)}{f'(\xi)}$. Therefore, the higher values of ACOC for $f_1(x)$ are caused by the fact that the second derivative of this function vanishes at the root. For $f_2(x)$ the computed ACOC agrees with the theoretical order of convergence of a method (if the latter converges). Finally, the equation with $f_3(x)$ has a multiple root and the convergence rate of all the methods is linear (if they converge). On the other hand, PM requires fewer iteration steps than Traub's method and has a larger domain of convergence than all the classical methods considered in this work.

3. GENERALIZATION OF PM TO SYSTEMS OF EQUATIONS

Below, method (4) is extended to systems of Eqs. (1). We show under which constraints on b and c (depending on α) the method has the third order of local convergence. Then the methods are compared as applied to a system of two nonlinear equations describing the motion of celestial bodies in the restricted four-body problem.

3.1. Convergence Analysis

To generalize method (4) to the multidimensional case, the iteration expression has to be rewritten so that in the denominator does not involve nonlinear function evaluations, since they become vectors in the multidimensional case. For this purpose, we assume that, at the first step, the iteration process $y_k = x_k - \alpha\frac{f(x_k)}{f'(x_k)}$, $f(x_k)$ can be expressed as $f(x_k) = \frac{1}{\alpha}(x_k - y_k)f'(x_k)$. Then, at the second step of method (4), the multiplier containing b and c can be rewritten as

$$\frac{f(x_k)^2}{bf(x_k)^2 + cf(y_k)^2} = \frac{1}{b + c\left(\frac{f(y_k)}{f(x_k)}\right)^2} = \frac{1}{b + c\left(\frac{af(y_k)}{(x_k - y_k)f'(x_k)}\right)^2} = \frac{1}{b + c\alpha^2\left[\frac{f[y_k, x_k]}{f'(x_k)}\right]^2}.$$

Now method (4) can be completely extended to systems of equations with several variables. The corresponding iterative procedure has the form of (5).

Before proving the order of convergence of the method in $\mathbb{R}^n$, we cite the following definition, which was given in [16]. The operator $[\cdot, \cdot, f] : D \times D \subset \mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathcal{L}(\mathbb{R}^n)$ is called the *first divided difference* of f if $[x, y; f](x - y) = f(x) - f(y)$, $\forall x, y \in D$.

Using the Hermite–Genocchi formula (see [17])

$$[x, x + h; f] = \int_0^1 f'(x + th)dt$$

and expanding $f'(x + th)$ in a Taylor series around the point x, we obtain

$$\int_0^1 f'(x + th)dt = f'(x) + \frac{1}{2}f''(x)h + \frac{1}{6}f'''(x)h^2 + O(h^3). \tag{8}$$

If $e = x - \xi$, assuming that $f'(\xi)$ is nonsingular yields

$$\begin{aligned} f(x) &= f'(\xi)[e + C_2e^2 + C_3e^3 + C_4e^4] + O(e^5), \\ f'(x) &= f'(\xi)[I + 2C_2e + 3C_3e^2 + 4C_4e^3] + O(e^4), \\ f''(x) &= f'(\xi)[2C_2 + 6C_3e + 12C_4e^2] + O(e^3), \\ f'''(x) &= f'(\xi)[6C_3 + 24C_4e] + O(e^2), \end{aligned} \tag{9}$$

where $C_k = (1/k!)[f'(\xi)]^{-1}f^{(k)}(\xi)$, $k = 2, 3, \ldots$.

Substituting (9) into (8) and setting $y = x + h$ and $e_y = y - \xi$, we obtain

$$[x, y; f] = f'(\xi)[I + C_2(e_y + e) + C_3e^2] + O(e^3). \tag{10}$$

Now we can formulate and prove the following result about nonlinear systems.

Theorem 2. *Let $f : D \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^n$ be a sufficiently differentiable function in an open neighborhood D of the solution ξ to the nonlinear system $f(x) = 0$. Assume that $f'(x)$ is nonsingular at ξ and $x^{(0)}$ is an initial approximation that is sufficiently close to ξ. Then the sequence $\{x^{(k)}\}_{k \geq 0}$ generated by iteration expression* (5) *converges to ξ with the order of convergence equal to* 3 *if $b = \frac{1 + \alpha^2}{2\alpha^2}$, $c = \frac{(1 + \alpha)}{2\alpha^2(\alpha - 1)}$, and $\alpha \notin \{0, 1\}$. Moreover, the error equation of the method has the form*

$$e^{(k+1)} = \left(-\frac{1}{2}(-2 + 2\alpha + \alpha^2 + \alpha^3)C_2^2 + 2(\alpha - 1)C_3\right)e^{(k)^3} + O(e^{(k)^4}),$$

where $C_k = (1/k!)[f'(\xi)]^{-1}f^{(k)}(\xi)$ for $k = 2, 3, \ldots$ and $e^{(k)} = x(k) - \xi$.

Proof. Expanding $f(x^{(k)})$ and $f'(x^{(k)})$ in Taylor series around ξ gives

$$f(x^{(k)}) = f'(\xi)[e^{(k)} + C_2e^{(k)^2} + C_3e^{(k)^3}] + O(e^{(k)^4}) \tag{11}$$

and

$$f'(x^{(k)}) = f'(\xi)[I + 2C_2e^{(k)} + 3C_3e^{(k)^2}] + O(e^{(k)^3}). \tag{12}$$

Moreover, assuming that $[f'(x^{(k)})]^{-1}f'(x^{(k)}) = f'(x^{(k)})[f'(x^{(k)})]^{-1} = I$, we obtain

$$[f'(x^{(k)})]^{-1} = [I + X_2e^{(k)} + X_3e^{(k)^2}][f'(\xi)]^{-1} + O(e^{(k)^3}), \tag{13}$$

where

$$X_2 = -2C_2, \quad X_3 = 4C_2^2 - 3C_3.$$

Then

$$[f'(x^{(k)})]^{-1}f(x^{(k)}) = e^{(k)} - C_2e^{(k)^2} + 2(-C_3 + C_2^2)e^{(k)^3} + O(e^{(k)^4}) \tag{14}$$

and the error equation at the first step becomes

$$e_y^{(k)} = y^{(k)} - \xi = (1-\alpha)e^{(k)} + \alpha C_2e^{(k)^2} - 2\alpha(-C_3 + C_2^2)e^{(k)^3} + O(e^{(k)^4}). \tag{15}$$

Now, using (10), we have

$$[y^{(k)}, x^{(k)}; f] = f'(\xi)[I + (2-\alpha)C_2e^{(k)} + \alpha C_2^2e^{(k)^2} + (3 - 3\alpha + \alpha^2)C_3e^{(k)^2}] + O(e^{(k)^3}). \tag{16}$$

Combining (13) with (16) gives

$$\frac{1}{\alpha}I - [f'(x^{(k)})]^{-1}[y^{(k)}, x^{(k)}; f] = \left(\frac{1}{\alpha} - 1\right)I + \alpha C_2e^{(k)} - (3\alpha C_2^2 + \alpha(\alpha - 3)C_3)e^{(k)^2} + O(e^{(k)^3}). \tag{17}$$

Thus, the Taylor expansion (around ξ) of the expression to be inverted at the second step of method (5) becomes

$$A = bI + c\alpha^2\left(\frac{1}{\alpha}I - [f'(x^{(k)})]^{-1}[y^{(k)}, x^{(k)}; f]\right)^2 = \left(b + \left(\frac{1}{\alpha} - 1\right)^2 c\alpha^2\right)I + 2c\alpha^3\left(\frac{1}{\alpha} - 1\right)C_2e^{(k)}$$

$$+ [c\alpha^2(\alpha^2 - 6 + 6\alpha)C_2^2 + (2c\alpha^3(\alpha - 3) - 2c\alpha^2(\alpha - 3))C_3]e^{(k)^2} + O(e^{(k)^3}).$$

Next, we define $A^{-1} = I + Y_1e^{(k)} + Y_2e^{(k)^2} + O(e^{(k)^3})$ so that $AA^{-1} = A^{-1}A = I$. Assume that the following expression is an identity matrix:

$$AA^{-1} = \left(b + \left(\frac{1}{\alpha} - 1\right)^2 c\alpha^2\right)I + \left[\left(b + \left(\frac{1}{\alpha} - 1\right)^2 c\alpha^2\right)Y_1 + 2c\alpha^3\left(\frac{1}{\alpha} - 1\right)C_2\right]e^{(k)}$$

$$+ \left[\left(b + \left(\frac{1}{\alpha} - 1\right)^2 c\alpha^2\right)Y_2 + 2c\alpha^3\left(\frac{1}{\alpha} - 1\right)C_2Y_1 + c\alpha^2(\alpha^2 - 6 + 6\alpha)C_2^2\right.$$

$$\left. + (2c\alpha^3(\alpha - 3) - 2c\alpha^2(\alpha - 3))C_3\right]e^{(k)^2} + O(e^{(k)^3}).$$

For this purpose, we need to satisfy the equations

$$b = 1 - \left(\frac{1}{\alpha} - 1\right)^2 c\alpha^2,$$

$$Y_1 = -2c\alpha^3\left(\frac{1}{\alpha} - 1\right)C_2,$$

$$Y_2 = -Y_1^2 - c\alpha^2(\alpha^2 - 6 + 6\alpha)C_2^2 - (2c\alpha^3(\alpha - 3) - 2c\alpha^2(\alpha - 3))C_3.$$

Then the error at the last iteration is given by

$$x^{(k+1)} - \xi = y^{(k)} - \xi - A^{-1}[f'(x^{(k)})]^{-1}f(y^{(k)}) = ((-1 + a)(-1 - \alpha - 2\alpha^2 c + 2\alpha^3 c)C_2)e^{(k)^2}$$

$$+ ((-2 + \alpha^2(4 - 8c) + 16\alpha^3 c - 12\alpha^6 c^2 + 4\alpha^7 c^2 + 3\alpha^5 c(-1 + 4c) - \alpha^4 c(5 + 4c))C_2^2$$

$$-(-1 + \alpha)(2 + 2\alpha - 8\alpha^3 c + 2\alpha^4 c + \alpha^2(-1 + 6c))C_3)e^{(k)^3} + O^{(k)^4}$$

and the third order of convergence is achieved only if $c = \dfrac{\alpha + 1}{2\alpha^2(\alpha - 1)}$ and $b = \dfrac{\alpha^2 + 1}{2\alpha^2}$. In this case, the error equation of the method has the form

$$e^{(k+1)} = \left(-\frac{1}{2}(-2 + 2\alpha + \alpha^2 + \alpha^3)C_2^2 + 2(\alpha - 1)C_3\right)e^{(k)^3} + O(e^{(k)^4}).$$

3.2. Application to Celestial Mechanics

As an example, we apply PM (5) to celestial mechanics, namely, to the Newtonian circular restricted four-body problem formulated on the basis of Lagrange triangular solutions [10, 11]. The nonlinear equations determining equilibrium solutions in the orbital plane of the bodies are

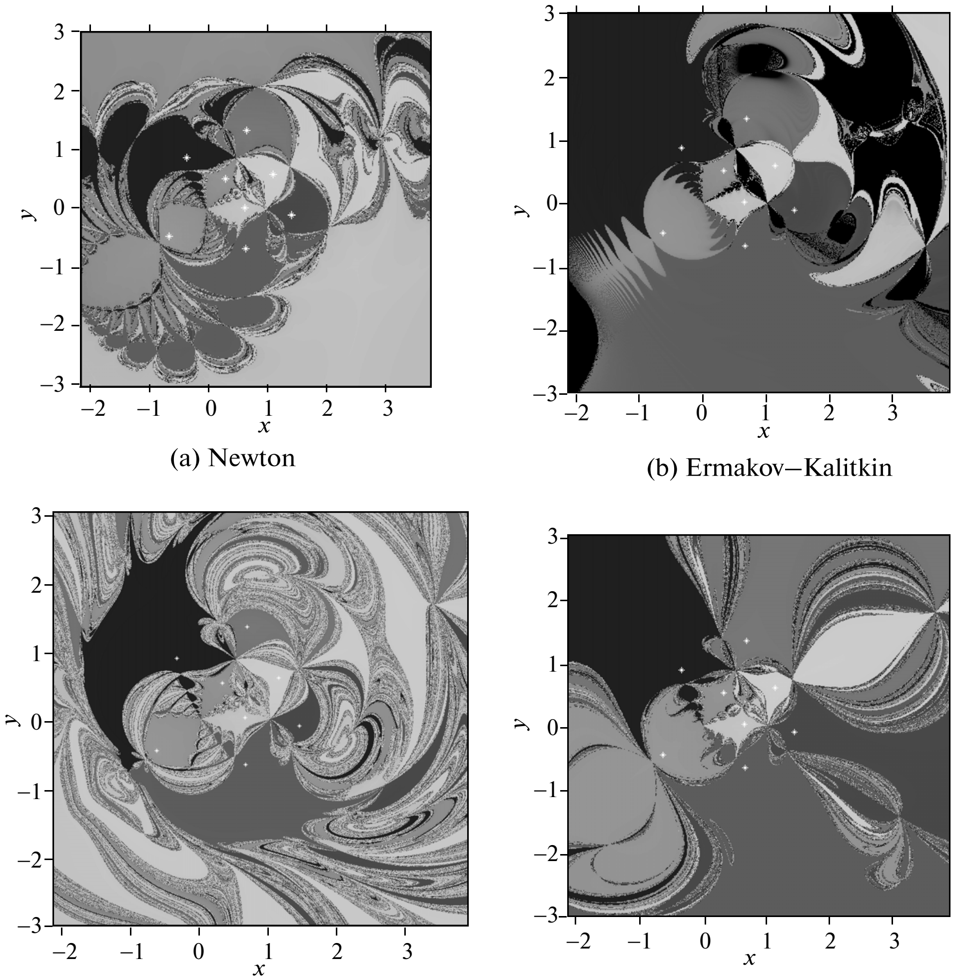
$$\begin{aligned}(\sqrt{3}x - y)\left(1 - \frac{1}{(x^2 + y^2)^{3/2}}\right) + \mu_1(\sqrt{3}(x - 1) + y)\left(1 - \frac{1}{((x - 1)^2 + y^2)^{3/2}}\right) &= 0,\\ 2y\left(1 - \frac{1}{(x^2 + y^2)^{3/2}}\right) + \mu_2(\sqrt{3}(x - 1) + y)\left(1 - \frac{1}{(1 - x + x^2 - \sqrt{3}y + y^2)^{3/2}}\right) &= 0\end{aligned} \tag{18}$$

and involve two mass parameters μ_1 and μ_2. For various values of $0 < \mu_1 < \mu_2 < 1$, system (18) has 8 to 10 solutions. As will be shown below, the basins of attraction for this system have chaotic, nonuniform, and nonsmooth boundaries with much noise. The solutions of system (18) can be computed in a more or less stable manner depending on the iterative method applied. Here, stability is understood in the sense that a small variation in the initial approximation for the iterative method can cause convergence to another solution. Accordingly, the problem of choosing a stable iterative scheme for solving nonlinear systems of type (18) is of interest, although efficient algorithms (based on Newton's method) for finding solutions of system (18) were developed in [18] and the bifurcation curve separating the domains with various numbers of solutions was constructed in the parameter plane (see [19]).

Let us analyze the dynamics of the compared iterative methods as applied to system (18). Some preliminaries from the theory of holomorphic dynamical systems are presented in Subsection 2.2. For $\mu_1 = 0.25$ and $\mu_2 = 0.35$, Fig. 7 shows the phase planes constructed for the iterative schemes under consideration. For these parameter values, system (18) has eight solutions (depicted by open circles), each having its own basin of attraction shown by a different color. Note that Ermakov–Kalitkin method (2), (3) produce fairly good results in the sense of boundary sharpness. However, in the upper right corner (Fig. 7b), we can see a rather large dark area corresponding to a divergence domain.

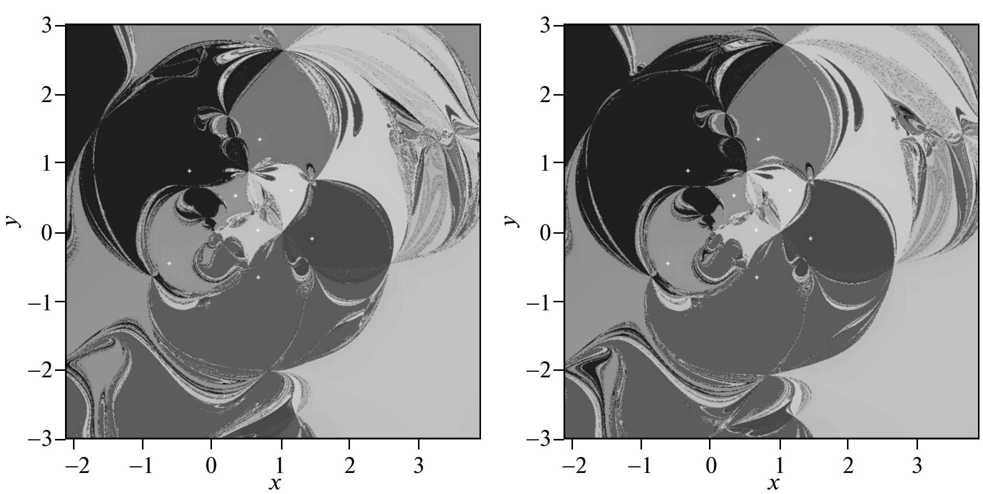
Figure 8 displays the phase planes for PM family (5) with small (in absolute value) parameters $\alpha = -0.01$ and $\alpha = 0.1$. It can be seen that method (5) is more stable than the other methods under consideration. Overall, for both equations and systems of equations, the PM family produces the best results for small (in absolute value) parameters α. This finding can be explained by the fact that, in the error equation for PM (see Theorems 1 and 2), the coefficient of e_k^3 depends, in fact, on α. Close-to-zero values of α decrease the coefficient of e_k^3, while accelerating and stabilizing PM.

Table 2 presents the numerical results obtained for system (18). Once again, five iterative schemes (Newton, Ermakov–Kalitkin, Traub, Jarratt, and PM (with $\alpha = 0.1$)) are compared in the case of various



(a) Newton (b) Ermakov–Kalitkin (c) Traub (d) Jarratt

Fig. 7. Phase planes for previously known methods as applied to system (18) with $\mu_1 = 0.25$ and $\mu_2 = 0.35$.



(a) $\alpha = -0.1$ (b) $\alpha = 0.01$

Fig. 8. Phase planes for the family of methods (5) as applied to system (18) with $\mu_1 = 0.25$ and $\mu_2 = 0.35$.

Table 2. Results of numerical tests for system (18)

M_1	M_2	$x^{(0)}$	Number of iterations				
			N	E–K	PM	T	J
0.25	0.35	(–0.2, –0.7)	18	15	12	17	7
		(3, 0.21)	15	15	13	24	13
		(3, –0.01)	14	14	13	14	23
0.1	0.2	(0.4, 0.8)	18	14	12	10	10
		(1, 1)	16	19	12	9	7
		(0.2, 3)	14	14	14	19	13

μ_1 and μ_2 and various initial approximations (denoted by $x^{(0)}$). The number of iterations required for achieving the root up to 10^{-2000} is also given in the table. Note that, with the initial data chosen, all the considered methods converged (see Table 2), though sometimes to different solutions. In this sense, Fig. 7c suggests that Traub's method is the most "unpredictable" as applied to system (18).

4. CONCLUSIONS

A new one-parameter family of third-order iterative methods was proposed for solving both nonlinear equations (4) and systems of equations (5). The main advantage of the methods is that they have a larger domain of convergence. By using the theory of holomorphic dynamical systems, the methods were compared with the Newton, Ermakov–Kalitkin, Traub, and Jarratt iterative schemes. The results of this comparison and a numerical experiment showed that the method proposed is superior in stability and the size of the convergence domain.

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